

COSC2671 Social Media and Network Analytics

Assignment 2 Worksheet

Group: Postgraduate Group 30

Student Names and Numbers: Dhanya Laxminarayan (s4139554); Irine Biju Panicker (s4137661); Aiswarya Sibi (s4133705)

1. Team Project Plan

Research Question: How do Reddit users interact in climate change discussions, and which users, topics, and subreddit communities act as hubs or bridges in the climate discourse network?

Dataset: cathw/reddit_climate_comment, a public Reddit climate discussion dataset hosted on HuggingFace.

Week Calculation: The project weeks are recorded using RMIT teaching weeks during the assignment period. Since the project was finalised around Week 12, the worksheet covers Week 8 to Week 12: proposal planning, analysis, report writing, final checking, and presentation preparation.

Planned Methodology:

- Sentiment analysis using VADER and Kruskal-Wallis testing
- TF-IDF keyword analysis and LDA topic modelling
- Directed weighted user interaction network construction
- PageRank, approximate bridge score, and community detection
- Subreddit similarity network and bipartite subreddit-topic network analysis

1.1 Weekly Milestones

Teaching Week	Milestone
Week 8	Dataset selection, research question, proposal planning, initial literature review
Week 9	Data loading, flattening, preprocessing, and initial sentiment analysis
Week 10	TF-IDF keyword analysis, LDA topic modelling, and topic interpretation
Week 11	User interaction network, PageRank, bridge score, community detection, and report writing
Week 12	Final report formatting, notebook checking, worksheet completion, and presentation preparation

1.2 Equal Contribution Statement

The workload was planned and reviewed as an equal contribution across the three team members. Each member contributed to both technical work and communication tasks, with different primary responsibilities but equal overall contribution.

Team Member	Student Number	Primary Contribution Area	Contribution %
Dhanya Laxminarayan	s4139554	Dataset exploration, preprocessing, sentiment analysis, report editing	33.33%
Irine Biju Panicker	s4137661	User interaction	33.33%

		network, PageRank, bridge score, visualisations	
Aiswarya Sibi	s4133705	LDA topic modelling, TF-IDF, subreddit networks, presentation preparation	33.33%
Total			100.00%

2. Weekly Timesheets

Each member maintained a weekly timesheet for the project period. Hours are balanced equally across members to reflect an even workload distribution.

Dhanya Laxminarayan

Teaching Week	Tasks Completed	Hours
Week 8	Explored candidate datasets, reviewed dataset fields, helped finalise research question	5
Week 9	Flattened dataset, cleaned text fields, prepared sentiment analysis workflow	6
Week 10	Ran VADER sentiment analysis, Kruskal-Wallis test, and helped interpret results	7
Week 11	Edited report sections, checked tables/figures, aligned results with notebook outputs	6
Week 12	Final proofreading, worksheet review, and presentation preparation	4
Total		28

Irine Biju Panicker

Teaching Week	Tasks Completed	Hours
Week 8	Compared datasets, checked network suitability, contributed to proposal planning	5
Week 9	Built user edge list from comment/reply relationships and cleaned network edges	6
Week 10	Computed PageRank, bridge score, and user interaction network measures	7
Week 11	Created network visualisations and interpreted influence/community findings	6
Week 12	Final code checking, presentation rehearsal, and worksheet preparation	4

Total		28
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Aiswarya Sibi

Teaching Week	Tasks Completed	Hours
Week 8	Reviewed related work, planned NLP methods, and helped define success criteria	5
Week 9	Prepared TF-IDF workflow and supported cleaning decisions for text modelling	6
Week 10	Ran LDA topic modelling, inspected topics, and labelled thematic frames	7
Week 11	Built subreddit similarity and bipartite subreddit-topic network outputs	6
Week 12	Prepared presentation content, checked references, and completed final review	4
Total		28

3. Individual Self-Reflection

Dhanya Laxminarayan: I contributed mainly to dataset exploration, preprocessing, sentiment analysis, and report editing. A challenge was selecting data that supported both NLP and real user interaction networks. The team worked collaboratively, and equal task tracking helped keep the workload balanced.

Irine Biju Panicker: I focused on user interaction network construction, PageRank, bridge score analysis, and visualisations. This project improved my understanding of how network measures can identify influential and connecting users. Better early planning helped the team divide the technical work more evenly.

Aiswarya Sibi: I worked mainly on TF-IDF, LDA topic modelling, subreddit networks, and presentation preparation. Interpreting noisy Reddit text was challenging, especially when assigning meaningful topic labels. The group communicated regularly and shared the workload equally across the project period.